Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third & Fourth Semester of B. Tech (EE) Examination

Apr-May 2018

EE201/EE201.01 Control Engineering

Date: 02.05.2018, Wednesday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q-1 (a) Define the following terms **[03]**

1. Control 2. System 3. Control system

(b) Explain closed loop control system by giving suitable example. Also enlist advantages for the same. **[07]**

Q-2 (a) Obtain the transfer function using Mason's Gain formulae for the signal flow graph shown in **Figure I.** **[05]**

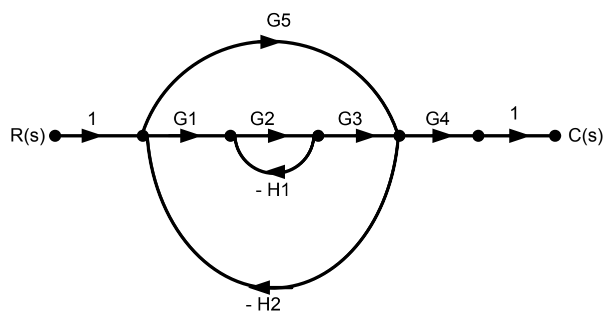

Figure – I

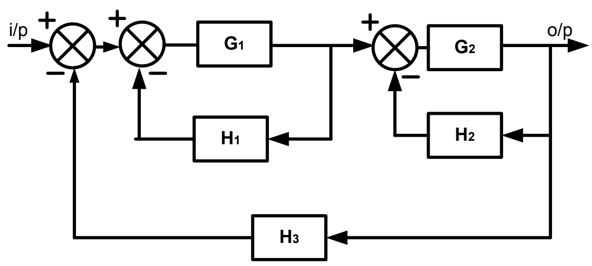

Figure - II

(b) Explain Missile launching system with the help of neat figure. **[05]**

OR

Q-2 (a) (i) Define following terms. **[03]**

Forward path, Source node, Sink node, Chain node, Feedback loop

(ii) Obtain T.F. for simple closed loop control system having forward path gain G(s), Feedback path Gain H(s). **[02]**

(b) Obtain Transfer Function using Block Diagram Reduction method for the block diagram shown in **Figure II.** **[05]**

Q-3 Attempt any three. **[15]**

(a) Explain the effects of PD controller in detail on 2^{nd} order control system.

(b) For a unity feedback system has $G(s) = \dfrac{40\,(s+2)}{s(s+1)(s+4)}$ Find

(i) Type & Order of the system (ii)Error constants (iii)Error for ramp input

(c) Explain in detail about various test signals used in control system. Also tabulate their values in Time domain and in Laplace domain.

(d) Draw a typical response c(t) with respect to time of 2^{nd} Order system and Define all the terms with their mathematical equations..

SECTION – II

Q-4 Obtain the transfer function from the given circuit shown in **Figure III.** **[04]**

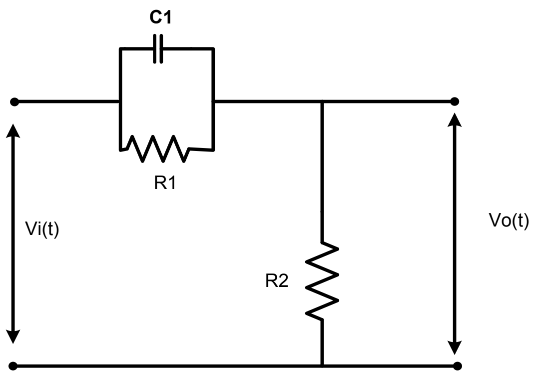

Figure – III

Q-5 (a) Draw the approximate root locus diagram for a closed loop control system whose open loop transfer function is given by **[07]**

$$G(s)H(s) = \frac{K}{s(s+5)(s+2)}$$. Comment on stability.

(b) The characteristics equation of the system is given by $s^6 + 4s^5 + 3s^4 - 16s^2 - 64s - 48 = 0$ **[06]**
Find the number of roots of these equations with positive real part, zero real part and negative real part.

OR

Q-5 (a) Sketch the root locus of the system on the graph paper with open loop transfer function of the system is given as $G(s)H(s) = \dfrac{K}{s\,(s+1)\,(s+2)\,(s+3)}$. **[07]**

Where K is varying from zero to infinity.

(b) The characteristics equation of the system is given by. **[06]**

I. $s^3 + 6s^2 + 11s + 6 = 0$

II. $s^3 + 4s^2 + s + 16 = 0$

Comment on stability using Routh - Hurwitz criterion.

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Q-6 Attempt any three. **[18]**

(a) A unity feedback system has

$$G(s) = \frac{2000}{s(s+1)(s+100)}.$$

Draw asymptotic Bode Plot. Determine Gain Margin.

(b) A unity feedback system has

$G(s) = \frac{10}{s(s+1)(s+2)}$. Comment on stability using Nyquist Plot.

(c) The Open Loop transfer function of unity Feed Back Control System is given by

$$G(s) = \frac{K}{s(1+0.001s)(1+0.25s)(1+0.1s)} .$$

Determine the value of K so that the system will have a phase margin of 40^0

What will be the gain margin? Use Bode Plot.

(d) Explain Phase Lead Compensation technique in detail.

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